

# **ASSIGNMENT 4: FROM GENE MUTATION TO DISEASE**

BIO 300 – Cell and Molecular Biology

Section B

## **Selected Study:**

*Investigating the Effect of an FMR1 Nonsense Mutation Associated with Fragile X Syndrome*

**Disease/Phenotype:** Fragile X Syndrome

**Gene:** *FMR1* — Fragile X Messenger Ribonucleoprotein 1

## **Submitted to:**

Lilibeth A. Bucol

## **Submitted by:**

Estrellanes, Klea Mae G.

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## Introduction

Changes in DNA sequence can affect biological function by changing the structure or production of proteins. A mutation can involve one or several nucleotides and lead to an amino-acid substitution, a premature stop codon, a frameshift, an insertion, a deletion or other molecular effects. An important link between genotype and phenotype is understanding how a change in DNA affects the coding sequence and predicted protein product.

Fragile X syndrome (FXS) is an inherited neurodevelopmental disorder linked to abnormalities in the FMR1 gene. It encodes fragile X messenger ribonucleoprotein (FMRP), an RNA-binding protein that controls messenger RNA transport and translation. Disruption of normal FMRP function may interfere with molecular processes critical for neuronal development and synaptic activity.

In this lab activity the NCBI RefSeq was used to obtain the normal coding sequence of FMR1, and it was analyzed in Galaxy for this lab activity. The coding sequence was manually rebuilt with a documented FMR1 nonsense variant, NM\_002024.6:c.80C>A. Wildtype and mutant sequences were translated with EMBOSS transeq and aligned with EMBOSS Needle. To demonstrate the differential effects of mutation type and position on the protein sequence, a second artificial three-nucleotide deletion was constructed.

The analysis was done based on the molecular relationship required in the activity:

DNA mutation -> coding-sequence change -> predicted protein change -> possible functional consequence -> disease phenotype.

The lab also notes that sequence translation predicts a protein sequence, but does not in and of itself prove actual protein expression, abundance, folding, localization or activity.

## A. Disease Background

Fragile X syndrome is a genetic disorder that affects brain development. Developmental delay, intellectual disability, learning difficulties, delayed speech and language development, attention problems, anxiety, hyperactivity, and autism-related behaviors are key features that may be present.

The disorder primarily affects the central nervous system, especially neurons and neuronal synapses. The molecular basis of classic Fragile X syndrome is most commonly expansion of the CGG trinucleotide repeat in the 5' untranslated region of FMR1. This can result in reduced or silenced normal gene expression.

The disease is tied to the X chromosome and is passed down in an X-linked way.

## Required Disease Questions

### 13. What is the disease or phenotype?

The disease under investigation is Fragile X syndrome (FXS) which is an inherited neurodevelopmental disorder that is characterized by loss of normal FMR1 function.

### 14. What are its major clinical characteristics?

Key clinical manifestations include developmental delay, intellectual disability, learning difficulties, speech and language delay, behavioral abnormalities, anxiety, attention problems and autism-associated behaviors.

### 15. Which cells, tissues, or organs are mainly affected?

The central nervous system is mainly affected, especially the neurons and neuronal synapses.

### 16. What is its genetic basis?

Fragile X syndrome is caused by pathogenic alterations in FMR1. The most common mechanism is expansion of the CGG repeats in the 5' untranslated region. This may lead to abnormal methylation and reduced or absent expression of FMR1.

### 17. What is its inheritance pattern, if applicable?

Fragile X syndrome is an X-linked condition.

Table 1. Summary of Fragile X syndrome.

| Information                    | Description                                                                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disease/phenotype              | Fragile X syndrome (FXS)                                                                                                                                                                  |
| Primary gene involved          | <i>FMRI</i>                                                                                                                                                                               |
| Major clinical characteristics | Developmental delay, intellectual disability, learning difficulties, speech and language delay, behavioral abnormalities, anxiety, attention problems, and autistic-like behaviors        |
| Major tissues/organs affected  | Central nervous system, particularly neurons and neuronal synapses                                                                                                                        |
| Genetic basis                  | Pathogenic alteration of <i>FMRI</i> . Classic FXS is commonly associated with expansion of CGG repeats in the 5' untranslated region and subsequent reduction or loss of FMRP expression |
| Inheritance                    | X-linked                                                                                                                                                                                  |

B. Gene and Normal Protein Function

The official gene symbol is *FMRI*, located on the X chromosome at Xq27.3. The gene encodes fragile X messenger ribonucleoprotein (FMRP).

FMRP functions as an RNA-binding regulatory protein. It participates in messenger RNA transport and translational regulation and is particularly important in neurons, where controlled protein synthesis contributes to synaptic development and plasticity.

FMRP is mainly associated with the cytoplasm, ribonucleoprotein complexes, and polyribosomes, although it can also participate in transport between the nucleus and cytoplasm.

Table 2. Reference information for the normal FMR1 gene and protein.

| Parameter             | Result                                  |
|-----------------------|-----------------------------------------|
| Official gene symbol  | <i>FMRI</i>                             |
| Gene name             | Fragile X messenger ribonucleoprotein 1 |
| Chromosomal location  | Xq27.3                                  |
| Reference transcript  | NM_002024.6                             |
| Reference protein     | NP_002015.1                             |
| CDS coordinates       | 262–2160                                |
| CDS length            | 1,899 bp                                |
| Normal protein length | 632 amino acids                         |
| Start codon           | ATG                                     |
| Stop codon            | TAA                                     |
| Reading frame         | +1                                      |
| First 10 amino acids  | MEELVVEVRG                              |
| Last 10 amino acids   | GQQPLVNGVP                              |

The NCBI reference record identifies NP\_002015.1 as a 632-amino-acid protein encoded by NM\_002024.6. The coding region corresponds to **NM\_002024.6:262..2160**.

Required Gene and Protein Questions

18. What is the official gene symbol?

*FMR1*

19. On which human chromosome is the gene located?

The gene is located on the **X chromosome** at **Xq27.3**.

20. What does the gene normally encode?

It encodes **fragile X messenger ribonucleoprotein (FMRP)**.

21. What is the normal biological function of the protein?

FMRP binds RNA and helps regulate mRNA transport, stability, and translation. These functions are especially important in neuronal development and synaptic plasticity.

22. Where in the cell is the protein normally found?

FMRP is mainly found in the cytoplasm, where it associates with ribonucleoprotein complexes and polyribosomes.

23. In what biological pathway or cellular process does it participate?

FMRP participates in **post-transcriptional regulation of gene expression**, mRNA transport, translational regulation, neuronal development, and synaptic plasticity.

C. Documented Mutation

The documented variant selected for analysis was:

**NM\_002024.6:c.80C>A**

This mutation changes **nucleotide 80** in the *FMR1* coding sequence from cytosine to adenine.

The normal codon at amino-acid position 27 is:

**TCA = Serine**

Following substitution:

**TAA = STOP**

The resulting predicted protein alteration is:

**p.Ser27Ter**

**Table 3. Documented FMR1 variant investigated in the activity.**

| Parameter               | Information                             |
|-------------------------|-----------------------------------------|
| Gene                    | <i>FMR1</i>                             |
| Reference transcript    | NM_002024.6                             |
| Variant                 | c.80C>A                                 |
| Nucleotide change       | C → A                                   |
| Codon change            | TCA → TAA                               |
| Protein consequence     | p.Ser27Ter (p.Ser27*)                   |
| Mutation type           | Nonsense single-nucleotide substitution |
| Amino-acid position     | 27                                      |
| Expected consequence    | Premature termination of translation    |
| ClinVar Variation ID    | 29987                                   |
| ClinVar accession       | VCV000029987.1                          |
| Clinical interpretation | Pathogenic                              |
| Supporting reference    | Grønskov et al. (2011)                  |

D. Mutation Hypothesis

Before generating the mutant protein, the **c.80C>A** substitution was predicted to change codon 27 from **TCA**, which encodes serine, to **TAA**, a stop codon.

Because the mutation is a nucleotide substitution rather than an insertion or deletion, the reading frame was predicted to remain unchanged. However, translation was expected to terminate prematurely at amino-acid position 27.

The mutant polypeptide was therefore predicted to contain only the first **26 amino acids** of a normally **632-amino-acid** FMRP protein. The resulting severely truncated product would lack most of the normal structural and functional regions of FMRP. The mutation affects one nucleotide, with cytosine at coding position 80 substituted by adenine.

## E. Methods

### 1. Retrieval of the Wild-Type Coding Sequence

The human *FMR1* transcript **NM\_002024.6** was obtained from NCBI RefSeq. The coding sequence corresponding to nucleotides **262–2160** was extracted in FASTA format.

The sequence was saved as:

**FMR1\_WT\_CDS.fasta**

The activity required use of the CDS for the translation experiment rather than the complete genomic gene or untranslated regions.

### 2. Galaxy History Preparation

A dedicated Galaxy history was created:

**Estrellanes\_FragileX\_FMR1\_Mutation\_Lab**

### 3. Translation of the Wild-Type CDS

The WT coding sequence was uploaded into Galaxy and translated using **EMBOSS transeq**.

The following settings were used:

- Reading frame: **Frame 1**
- Genetic code: **Standard**
- Terminal stop removal: **Yes**

The resulting predicted WT protein was saved as:

**FMR1\_WT\_protein.fasta**

### 4. Creation of the Documented Mutant

A copy of the WT coding sequence was produced so the original WT sequence remained unchanged.

Coding nucleotide 80 was changed from:

**C → A**

The local WT sequence:

**ATGAAGATTCAATAAC...**

was changed to:

**ATGAAGATTAAATAAC...**

This altered codon 27 from:

**TCA → TAA**

The mutant sequence was saved as:

**FMR1\_c80C-A\_mutant\_CDS.fasta**

**Number of bases substituted: 1**

**Mutation type:** Nonsense single-nucleotide substitution

### 5. Translation of the Documented Mutant

The documented mutant CDS was translated using EMBOSS transeq with the same settings used for the WT sequence.

The output was saved as:

**FMR1\_c80C-A\_mutant\_protein.fasta**

**6. WT–Mutant Protein Alignment**

The WT and mutant computational protein sequences were aligned using **EMBOSS Needle**.

The following were recorded:

- sequence identity,
- sequence similarity,
- alignment gaps,
- first amino-acid difference,
- and mutation position.

**7. Artificial Mutation Experiment**

A separate copy of the original WT coding sequence was used for the second experiment.

Exactly three nucleotides, **GGC**, corresponding to codon 10, were deleted.

The WT beginning:

**ATG GAG GAG CTG GTG GTG GAA GTG CGG GGC TCC...**

became:

**ATG GAG GAG CTG GTG GTG GAA GTG CGG TCC...**

The artificial sequence was saved as:

**FMR1\_artificial\_3bp\_deletion\_CDS.fasta**

It was translated using EMBOSS transeq with the same translation parameters.

The output was saved as:

**FMR1\_artificial\_3bp\_deletion\_protein.fasta**

**F. Results**

The WT *FMR1* CDS contained **1,899 nucleotides** and produced a predicted protein sequence of **632 amino acids**.

**Table 4. Wild-Type Results**

| Parameter                | Result     |
|--------------------------|------------|
| CDS length               | 1,899 bp   |
| Predicted protein length | 632 aa     |
| Reading frame            | +1         |
| Start codon              | ATG        |
| Stop codon               | TAA        |
| First 10 amino acids     | MEELVVEVRG |
| Last 10 amino acids      | GQQPLVNGVP |

The documented mutant retained the same CDS length because the mutation involved substitution of one nucleotide rather than insertion or deletion. The Galaxy-predicted WT protein was consistent with the

accepted NCBI reference protein NP\_002015.1, including the expected 632-amino-acid length and terminal amino-acid sequences.

Table 5. Documented Mutant Results

| Parameter                                 | Result      |
|-------------------------------------------|-------------|
| CDS length                                | 1,899 bp    |
| Mutation                                  | c.80C>A     |
| Codon change                              | TCA → TAA   |
| First amino-acid difference               | Position 27 |
| WT amino acid                             | Serine (S)  |
| Mutant result                             | STOP (*)    |
| Reading-frame change                      | No          |
| Premature stop                            | Yes         |
| Predicted translated residues before stop | 26 aa       |

Approximate amino acids affected: 606 residues of the normal 632-aa protein are absent from the predicted truncated product

The WT translation contained:

...KDVHEDSITV...

while the documented mutant contained:

...KDVHED\*ITV...

The appearance of the \* at position 27 confirmed generation of the expected premature stop codon.

G. WT versus Mutant Protein Comparison

EMBOSS Needle was used to compare the full computational translations.

Table 6. EMBOSS Needle Results

| Parameter        | Result          |
|------------------|-----------------|
| Alignment length | 632             |
| Identity         | 631/632 (99.8%) |
| Similarity       | 631/632 (99.8%) |
| Gaps             | 0/632 (0.0%)    |
| First difference | Position 27     |
| Sequence change  | S → *           |
| Score            | 3296.0          |

Interpretation of the Alignment

The **99.8% identity** reported by Needle must be interpreted carefully. EMBOSS transeq continued displaying amino acids downstream of the internal \*, allowing the program to align the remaining codons.

Biologically, the **TAA codon at position 27 represents termination of translation**. Therefore, the high numerical sequence identity does not mean the mutant biological protein would retain 99.8% of normal FMRP structure.

Required Comparison Questions

24. At what amino-acid position do the sequences first differ?

The sequences first differ at **amino acid position 27**.

**25. Is only one amino acid affected?**

At the computational alignment level, the direct difference occurs at position 27. However, biologically, the premature stop codon prevents normal translation of the downstream portion of the protein.

**26. Are multiple downstream amino acids changed?**

The downstream codons were not altered by the nucleotide substitution, but they would not normally be translated after the premature stop codon.

**27. Was an amino acid deleted or inserted?**

No. No amino acid was inserted or deleted. A serine codon was converted into a stop codon.

**28. Was a premature stop codon produced?**

Yes. A premature termination codon was produced at **position 27**.

**29. Did the reading frame change?**

No. The reading frame remained unchanged.

**30. Did the protein length change?**

Yes. The WT reference protein contains **632 amino acids**, while the documented mutant is predicted to terminate after **26 translated amino acids**.

**31. Is the mutation missense, nonsense, frameshift, in-frame deletion/insertion, repeat expansion, or another type?**

The mutation is a **nonsense single-nucleotide substitution**.

**H. Artificial Mutation Experiment**

Before translation, deletion of the three-nucleotide codon GGC was predicted to remove one glycine without shifting the reading frame. The artificial mutant CDS was predicted to decrease from 1,899 bp to 1,896 bp, while the predicted protein length was expected to decrease from 632 to 631 amino acids. No premature stop codon was expected.

The artificial mutation involved deletion of the three nucleotides:

**GGC**

Because three nucleotides correspond to one complete codon, the reading frame remained unchanged.

The WT predicted protein began:

**MEELVVEVRGSNGAF...**

while the artificial mutant began:

**MEELVVEVRSNGAF...**

The glycine at amino-acid position 10 was absent.

Table 7. Artificial Mutation Results

| Parameter              | WT       | Artificial Mutant |
|------------------------|----------|-------------------|
| CDS length             | 1,899 bp | 1,896 bp          |
| Protein length         | 632 aa   | 631 aa            |
| Nucleotides deleted    | —        | GGC               |
| Protein change         | —        | p.Gly10del        |
| Mutation type          | —        | In-frame deletion |
| Reading frame changed? | No       | No                |
| Premature stop codon   | No       | No                |

The experiment therefore confirmed the prediction that deletion of exactly three nucleotides can remove one amino acid without disrupting the downstream reading frame.

**I. Comparison of the Documented and Artificial Mutations**

Table 8. Comparison of WT and the Two Mutants

| Feature | WT | Documented Mutation | Artificial Mutation |
|---------|----|---------------------|---------------------|
|---------|----|---------------------|---------------------|

|                                 |             |                                                                        |                                                          |
|---------------------------------|-------------|------------------------------------------------------------------------|----------------------------------------------------------|
| CDS length                      | 1,899 bp    | 1,899 bp                                                               | 1,896 bp                                                 |
| Protein length                  | 632 aa      | Predicted termination after 26 aa                                      | 631 aa                                                   |
| DNA change                      | None        | c.80C>A                                                                | GGC deletion                                             |
| Mutation type                   | WT          | Nonsense substitution                                                  | In-frame deletion                                        |
| Reading frame changed?          | No          | No                                                                     | No                                                       |
| Premature stop?                 | No          | Yes                                                                    | No                                                       |
| Protein effect                  | Normal      | p. Ser27Ter                                                            | p. Gly10del                                              |
| Main consequence                | Normal FMRP | Premature translation termination                                      | Loss of one amino acid                                   |
| Amino acids affected            | None        | Ser27 becomes STOP; ~606 normal residues absent from predicted product | Gly10 deleted                                            |
| Expected functional consequence | Normal FMRP | Severe loss of normal FMRP function predicted                          | Functional effect uncertain from sequence analysis alone |

The documented mutation and artificial mutation produced different effects even though neither caused a frameshift. The documented mutation changed a sense codon into a stop codon, whereas the artificial mutation removed one complete codon.

J. Molecular Interpretation

The documented mutation can be represented as:

**FMR1 c.80C>A**  
↓  
**C → A substitution**  
↓  
**TCA → TAA**  
↓  
**Ser27 → STOP**  
↓  
**Premature termination of translation**  
↓  
**Severely truncated predicted FMRP**  
↓  
**Loss of normal RNA-regulatory functions**  
↓  
**Disturbance of neuronal and synaptic processes**  
↓  
**Fragile X syndrome phenotype**

The computational analysis directly supports the nucleotide change, codon change, presence of the premature stop signal, and predicted sequence consequence.

However, actual protein abundance, stability, localization, neuronal physiology, and clinical effects cannot be demonstrated by sequence translation alone.

K. Interpretation Questions

32. Why does the exact location of a mutation matter?

The exact location matters because different regions of a gene encode different functional portions of a protein. A mutation near the beginning of a coding sequence may have a major effect if it introduces an early stop codon. In the documented *FMRI* mutation, termination occurs at amino acid 27 of a normally 632-amino-acid protein.

33. Why can deleting three nucleotides produce a different result from deleting one or two nucleotides?

Codons consist of three nucleotides. Deleting three nucleotides can remove one complete codon while preserving the downstream reading frame. Deleting one or two nucleotides generally shifts the reading frame and changes many downstream codons.

**34. Does every mutation change the amino-acid sequence? Explain.**

No. Some nucleotide substitutions are synonymous because different codons can encode the same amino acid. Mutations in noncoding regions may also affect gene regulation without changing the protein sequence.

**35. Does every amino-acid substitution destroy protein function? Explain.**

No. The effect depends on the position and biochemical importance of the affected amino acid. Some substitutions have little effect, while others can disrupt folding, binding, stability, or biological activity.

**36. Why can a frameshift affect many amino acids even if only one nucleotide was deleted?**

Deleting one nucleotide changes the grouping of downstream nucleotides into codons. Consequently, many subsequent codons and amino acids may change, and a premature stop codon may eventually appear.

**37. Why might a premature stop codon produce a nonfunctional protein?**

A premature stop codon terminates translation before the complete protein is produced. The resulting truncated protein may lack domains required for normal structure and biological function.

**38. Could a mutation affect protein function without greatly changing protein length?**

Yes. A mutation affecting only one amino acid can disrupt an important structural, binding, catalytic, or regulatory region even when the overall protein length remains almost unchanged.

**39. Could a mutation cause disease without changing the protein sequence? Give a possible molecular mechanism.**

Yes. A mutation in a promoter, enhancer, untranslated region, or other regulatory sequence may alter gene expression without directly changing the encoded amino-acid sequence. Fragile X syndrome provides an example because abnormal CGG-repeat expansion in the *FMRI* 5' UTR can interfere with normal gene expression.

**40. What evidence from your analysis supports the proposed molecular mechanism of your disease?**

The analysis demonstrated that **c.80C>A** changed codon 27 from **TCA to TAA**. Galaxy translation produced a stop symbol at amino acid 27, supporting the predicted premature termination of FMRP translation.

**41. Which conclusions are supported directly by your computational results, and which require evidence from published experimental studies?**

The computational results directly support the nucleotide substitution, codon change, reading-frame status, premature stop codon, and predicted amino-acid sequence effect. Conclusions regarding actual FMRP expression, degradation, localization, neuronal dysfunction, and the clinical phenotype require experimental and published biological evidence.

## **L. Limitations**

This investigation had several limitations. First, the activity was mainly a **computational sequence-analysis experiment**. EMBOSS transeq predicted the amino-acid sequences encoded by the wild-type and mutant coding sequences but did not experimentally measure FMRP production, stability, folding, intracellular localization, or biological activity.

Second, EMBOSS transeq continued displaying translation downstream of the internal stop codon. Consequently, EMBOSS Needle reported an alignment identity of 99.8%. This does not indicate that the mutant biological protein would be almost identical to normal FMRP because cellular translation is expected to terminate at the premature stop codon.

Third, the documented **c.80C>A** variant represents a coding mutation selected for sequence analysis. The most common molecular mechanism of classic Fragile X syndrome involves CGG-repeat expansion and altered *FMRI* expression rather than this particular coding substitution.

Finally, the artificial **p. Gly10del** mutation was designed as a controlled sequence experiment. Its biological effect was not experimentally tested; therefore, only its sequence-level consequence can be concluded from this activity.

## **M. Conclusion**

This laboratory activity demonstrated how different DNA mutations can produce distinct consequences at the protein level. The documented *FMRI* **c.80C>A** mutation changed codon 27 from **TCA**, which encodes serine, to **TAA**, a premature stop codon. Galaxy translation confirmed the presence of the stop signal at amino-acid position 27, producing the nonsense variant **p. Ser27Ter**.

Although the mutation did not alter the reading frame or CDS length, it was predicted to terminate protein synthesis after only 26 amino acids instead of producing the normal 632-amino-acid FMRP.

In contrast, deletion of the three-nucleotide codon **GGC** produced an **in-frame p. Gly10del mutation**. The artificial mutant protein contained 631 amino acids and maintained the downstream reading frame.

These results demonstrate that the molecular effect of a mutation depends not only on the number of nucleotides altered but also on their **exact location, effect on codon identity, and relationship to the reading frame**. The activity also demonstrated the importance of distinguishing predicted sequence consequences from experimentally verified biological effects.

N. References

**Grønskov, K., Brøndum-Nielsen, K., Dedic, A., & Hjalgrim, H. (2011).** A nonsense mutation in *FMR1* causing fragile X syndrome. *European Journal of Human Genetics*, 19(4), 489–491.  
<https://doi.org/10.1038/ejhg.2010.223>

**Hunter, J. E., Berry-Kravis, E., Hipp, H., & Todd, P. K. (2024).** *FMR1 Disorders*. GeneReviews®. National Center for Biotechnology Information. <https://www.ncbi.nlm.nih.gov/books/NBK1384/>

**National Center for Biotechnology Information. (2026).** *FMR1 fragile X messenger ribonucleoprotein 1 [Homo sapiens]*. NCBI Gene. <https://www.ncbi.nlm.nih.gov/gene/2332>

**National Center for Biotechnology Information. (2026).** *NM\_002024.6: Homo sapiens FMR1 transcript variant ISO1, mRNA*. NCBI RefSeq. [https://www.ncbi.nlm.nih.gov/nucore/NM\\_002024.6](https://www.ncbi.nlm.nih.gov/nucore/NM_002024.6)

**National Center for Biotechnology Information. (2026).** *NP\_002015.1: Fragile X messenger ribonucleoprotein 1 isoform ISO1 [Homo sapiens]*. NCBI RefSeq. [https://www.ncbi.nlm.nih.gov/protein/NP\\_002015.1](https://www.ncbi.nlm.nih.gov/protein/NP_002015.1)

**National Center for Biotechnology Information. (2026).** *ClinVar: NM\_002024.6(FMR1):c.80C>A (p.Ser27Ter)*. Variation ID 29987, VCV000029987.1. <https://www.ncbi.nlm.nih.gov/clinvar/variation/29987/>

Appendix A. Galaxy History Evidence

Galaxy

Activities

Help Updated

Upload

Tools

Galaxy2

Workflows

Interactive Tools

Visualization

More

Galaxy is an open source, web-based platform for data intensive biomedical research. If you are new to Galaxy [start here](#) or consult our help resources. You can install your own Galaxy by following the [tutorial](#) and choose from thousands of tools from the [Tool Shed](#). New to Galaxy? Watch "What is Galaxy?" below.

JXTX

James P. Taylor Foundation

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Latest from BioRxiv

Introduction to the Data Analysis Platform Galaxy (in German) • Freiburg, Germany

Sep 10, 2024

The Image Analysis in Galaxy community is happy to share the first publication on bioRxiv.

Sep 10, 2024

at 3:50:22 in Freiburg, Minu Hagar Anant presented how reproducible, multi-layer management.

View more on BioRxiv →

History

search datasets

Estrellanes\_FragileX\_FMR1\_Mutation\_Lab

Default Storage

12.4 GB

7: FMR1\_artificial\_stop\_deletion\_protein.fasta

6: FMR1\_artificial\_stop\_deletion\_CDS.fasta

5: needle on dataset 2 and 4

4: FMR1\_c80C-A\_mutant\_protein.fasta

3: FMR1\_c80C-A\_mutant\_CDS.fasta

2: FMR1\_WT\_protein.fasta

1: FMR1\_WT\_CDS.fasta

2: FMR1\_WT\_protein.fasta

ok

Preview

Visualize

Details

Edit

>262-2160.1 Homo sapiens fragile X messenger ribonucleoprotein 1 (FMR1), transcript variant ISO1, mRNA

MEELWVEVRGSGAFYKAFVKDVHEDSITVAFENIMQDPDRITPFHDVRFPPPGVYNKDIN

ESDEVVYSRANEKEPCCHMLAKVRMIKGEFYIYAACDATYNEIVTIERLSRVNPNKP

ATKDTFHKIKLDVPEDLRQMKAKEAAHKDFKAVGAFSVTYDPENYQLVILSINEVTSKR

AHMLIDMHFRSLRTKLSLIMRNEEASQLESSRQLASRFHEQFTVREDLMGLAIGTHGAN

IQQARKVPGVTAIDLDEDTCTFHIYGEDQDAVKKARSFLEFAEDVIQVPRNLVGKVIQKN

GKLIQETVDKSGVVRVRIEAEENEKNVPQEEIMPNNLSPSNNSRVGPNAPEEKKHLDIKE

NSTHFSQPNSTKVQVRVLVASSVAGESQKPELKAMQGMVPFVFGTKDSIANATVLLDYH

LNLYKEVDQLRLRLQIDEQLRQIGASSRPPNIRTDKEKSVYTDGQGMGRGSRPYRNRG

HGRRGPGYTSGTNSEASNASETSDHREDLSQNSLAPTEEERESFLRRGDGRRRGGGGRRG

QGGRRGGGFGKNDHDSRTDNRPNRPREAKGRTTDSLSQIRVDCNNERSVHTKTLQNTSS

EGSRLRTGKDRNQKKEKPDSDVGQQLVNGVP

4: FMR1\_c80C-A\_mutant\_protein.fasta

ok

Preview

Visualize

Details

Edit

>262-2160.1 Homo sapiens fragile X messenger ribonucleoprotein 1 (FMR1), transcript variant ISO1, mRNA

MEELWVEVRGSGAFYKAFVKDVHEDSITVAFENIMQDPDRITPFHDVRFPPPGVYNKDIN

ESDEVVYSRANEKEPCCHMLAKVRMIKGEFYIYAACDATYNEIVTIERLSRVNPNKP

ATKDTFHKIKLDVPEDLRQMKAKEAAHKDFKAVGAFSVTYDPENYQLVILSINEVTSKR

AHMLIDMHFRSLRTKLSLIMRNEEASQLESSRQLASRFHEQFTVREDLMGLAIGTHGAN

IQQARKVPGVTAIDLDEDTCTFHIYGEDQDAVKKARSFLEFAEDVIQVPRNLVGKVIQKN

GKLIQETVDKSGVVRVRIEAEENEKNVPQEEIMPNNLSPSNNSRVGPNAPEEKKHLDIKE

NSTHFSQPNSTKVQVRVLVASSVAGESQKPELKAMQGMVPFVFGTKDSIANATVLLDYH

LNLYKEVDQLRLRLQIDEQLRQIGASSRPPNIRTDKEKSVYTDGQGMGRGSRPYRNRG

HGRRGPGYTSGTNSEASNASETSDHREDLSQNSLAPTEEERESFLRRGDGRRRGGGGRRG

QGGRRGGGFGKNDHDSRTDNRPNRPREAKGRTTDSLSQIRVDCNNERSVHTKTLQNTSS

EGSRLRTGKDRNQKKEKPDSDVGQQLVNGVP



**GITHUB REPOSITORY LINK:** <https://github.com/kleamae08/FragileX-FMR1-Mutation-Lab.git>